

Same product, same outward flux

EXACT SIGN/PRODUCT DIAGRAM

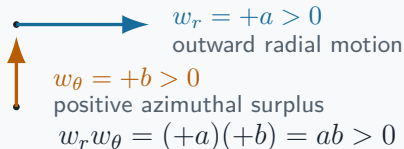
A local identity for the added pulse velocity w at one fixed cylindrical radius $r > 0$

Orientation key only $+e_r$: away from the axis

$+e_\theta$: chosen positive tangent direction

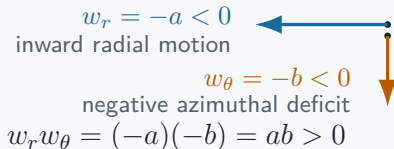


Case A: outward + surplus



$r w_r w_\theta = rab > 0 \quad (r > 0)$
positive outward angular-momentum flux factor

Case B: inward + deficit



$r w_r w_\theta = rab > 0 \quad (r > 0)$
positive outward angular-momentum flux factor

Sign reversal leaves the product fixed.

$$(w_r, w_\theta) \mapsto (-w_r, -w_\theta)$$

$$(-w_r)(-w_\theta) = w_r w_\theta.$$

Covariance bridge. $C_{r\theta}(w) = \langle\langle w_r w_\theta \rangle_\theta\rangle_Y.$

Fixed-amplitude shared cosine phase: $\langle w_r w_\theta \rangle_\theta = ab \langle \cos^2 \phi \rangle_\theta = ab/2 > 0.$

The angular frequency is a nonzero integer.

Scope and limit. Arrows show component sign and orientation only, not pulse geometry, paths, profiles, magnitudes, support, or the full three-dimensional solution. Their length and placement carry no scale or geometric information.
EE-ILL-P006-001 Source: paper p. 6; equation (7.23); Proposition 7.5